

Physics 390 In-Class Activity 13***MRI SLICE SELECTION***

$$\nu(z) = \bar{\gamma}B(z) = \bar{\gamma}(B_0 + G_z z) = \nu_L + \bar{\gamma}G_z z$$

The equation above describes how the resonance excitation frequency varies with z in a linear field gradient with strength G_z . Here $\bar{\gamma} = 42.5800 \text{ MHz/T}$.

- (A) Assume that the main magnetic field strength is $B_0 = 1.5000 \text{ T}$. Show that the resonance frequency at $z = 0$ ($\nu_L = \bar{\gamma}B_0$) is 63.8700 Hz .

$$\nu_L = \bar{\gamma}B_0 = 42.5800 \times 1.5000 = 63.8700 \text{ MHz}.$$

- (B) Assume that a range of frequencies $63.9000 \pm 0.0002 \text{ MHz}$ is used to excite protons in the patient. Also assume that the gradient strength $G_z = 10^{-2} \text{ T/m}$. What is the thickness of the excited slice?

$$\Delta\nu = \bar{\gamma}G_z\Delta z$$

$$\Delta z = \frac{\Delta\nu}{\bar{\gamma}G_z} = \frac{0.0004}{42.5800 \times 0.01} = .00094 \text{ m} = 0.94 \text{ mm}$$

- (C) Determine the z location of the slice. (Hint: The z location is determined by the center frequency 63.9000 .)

$$\nu(z) = \nu_L + \bar{\gamma}G_z z$$

$$63.9000 = 63.8700 + \bar{\gamma}G_z z$$

$$z = \frac{0.0300}{42.5800 \times 0.01} = 0.07 \text{ m} = 7 \text{ cm}.$$